8 Resonance

INTRODUCTION

The study of resonance is very beneficial in the area of communication. There are many things which are crucially based on the principle of resonance. For instance, the ability of a radio receiver to select a certain frequency transmitted by a transmitting station and to eliminate frequencies from other transmitting stations.

A two-terminal network offers a complex impedance consisting of resistive and reactive components. If a sinusoidal voltage is applied to such a network, the current is in phase with the voltage under some conditions, the impedance offered by the network is purely resistive. This phenomenon is termed as resonance and the frequency of the applied signal at which the resonance occurs is known as the resonant frequency.

Grey Matter Alert!

Heinrich Hertz, the German physicist published the first example of an electrical resonance curve in the year 1887 in his groundbreaking paper on the discovery of radio waves, showing the length of the spark obtained from his spark-gap LC resonator detectors as a function of frequency.

GENERAL DEFINITION

- When any quantity or a system oscillates between two forms, then it is said to be in resonance.
- In electrical networks, energy oscillates between electrostatic and electromagnetic forms.
- If energy keeps getting exchanged between L and C, then the network does not draw any energy from the source for L and C but only draws for R. So, source supplies the energy being lost in R, whereas energy stored in L and C remains constant.

  For the source, it appears that the network is resistive.

  $\therefore$ Power factor ($\cos\theta$) = 1, and the source only supply real power.

  Since the source only observes resistance.

  $\therefore$ Imaginary(z) = 0 at source terminals.

Types of resonance:

1) Series resonance

2) Parallel resonance

A circuit is said to be at resonance when source voltage and source current are in phase. So, phase angle $\phi = 0$, the power factor of the circuit = 1 (unity power factor).

- If a sinusoidal voltage is applied to an ac network, the impedance offered by the network is purely resistive under certain circumstances.
- A resonance circuit must have an inductance and a capacitance. The resistance will be present either due to the lack of ideal elements or due to the presence of a resistive element itself.
- When resonance occurs, the energy absorbed by one reactive element equals the energy emitted by another reactive element in the system at any given time.

The total energy of reactive elements = constant

- The resonant condition in the AC circuit may be achieved by varying the frequency of the supply, keeping the network elements constant or by varying inductance (L) or capacitance (C) keeping the frequency constant.
- Resonance is the frequency response of a circuit when it is operating at its natural frequency.
- The entire network behaves as purely resistive at resonance, and maximum power is drawn from the source.
- At resonance, the natural frequency of the network matches the supply frequency.

Natural frequency vs supply frequency:

- Natural frequency refers to the frequency at which there can be self-sustained oscillation i.e. oscillations that do not die out (undamped).
- When we apply the same frequency source then the condition is called resonance and the network starts to exhibit self-sustained oscillation.

QUALITY FACTOR

- To undergo a good observable resonance practically we need good quality energy storing components L and C. Thus, quality is measured as the figure of merit or quality factor given by

$$Q = 2\pi \times \frac{\text{Maximum energy stored per cycle of supply}}{\text{Energy dissipated per cycle of supply}}$$



Rack Your Brain

What is the condition of resonance in the RLC circuit?

- In the practical application of the resonance phenomenon, the minimum quality factor to be considered is 10 but in application to the power system, etc. the quality factor can be as high as 100.

The quality factor of different elements:

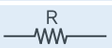
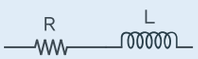
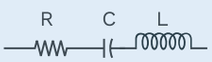
Element	Q Factor
	0
	∞
	∞
	$\frac{\omega L}{R} = \tan\theta$
	$\frac{1}{\omega RC}$
	$\frac{1}{R}\sqrt{\frac{L}{C}}$ (under resonance)

Table 8.1

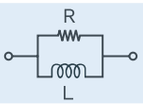
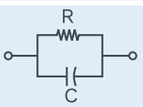
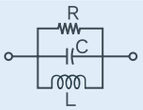
No need of resonance	
	$\frac{R}{\omega L}$
	ωRC
	$R\sqrt{\frac{C}{L}}$ (under resonance)

Table 8.2

- To determine quality factor,

$$Q = \tan\theta = \frac{\text{reactive power}}{\text{real power}}$$

Grey Matter Alert!

The quality factors of the network is a measure of the sharpness of the curve for the current.

Series RL circuit:

$i = I_m \sin \omega t$

maximum energy stored = $\frac{1}{2} L I_m^2$

power dissipated = $I_{rms}^2 R$

energy dissipated per cycle = $\left(I_{rms}^2 R\right) T$

Quality factor,

$$Q = 2\pi \times \frac{\frac{1}{2} L I_m^2}{I_{rms}^2 R\, T} = \frac{2\pi}{T} \frac{\left(\frac{I_m}{\sqrt{2}}\right)^2 L}{R I_{rms}^2} = \frac{\omega L}{R}$$

Direct method:

$$Q = \tan\theta = \frac{\text{reactive power}}{\text{real power}} = \frac{I^2 X}{I^2 R}$$

$$Q_L = \frac{X}{R} = \frac{\omega L}{R}$$

e.g. parallel RC

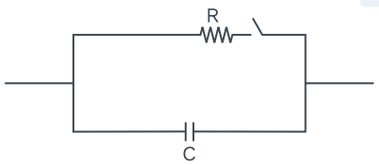


Fig. 8.1

$$Q = \frac{\text{reactive power}}{\text{real power}} = \frac{\frac{V^2}{X}}{\frac{V^2}{R}} = \frac{R}{X}$$

$$Q = \frac{R}{\frac{1}{\omega C}} = \omega C R$$

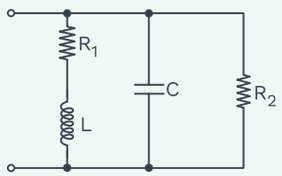
Note:

1) For parallel circuits, use voltage and in series circuit, use current.

2) In the RLC circuit, we can compute quality factors using L as well as C.

Previous Years' Question

The half-power bandwidth of the resonant circuit of the figure can be increased by:



I) Increasing R_1
II) Decreasing R_1
III) Increasing R_2
IV) Decreasing R_2

Grey Matter Alert!

For a series R-L-C circuit, quality factor represents the voltage amplification factor and must have a high value for any tuned circuit.

SERIES RESONANCE

General Series resonant circuit consisting of ideal elements R, L, and C are shown in the figure. When the circuit's resulting reactance is zero, the circuit is said to be in resonance.

$Z = R + j(X_L - X_C)$

At resonance Vs and I are in phase.

So, $Z = R$

So net reactance $X_L - X_C = 0$

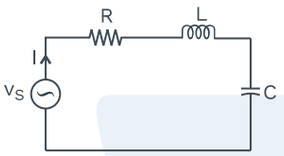


Fig. 8.2 Condition for Resonance Imaginary(z) = 0

$$\omega^2 = \frac{1}{LC}$$

$$\Rightarrow \omega = \frac{1}{\sqrt{LC}} \text{ rad/s}$$

ω = source frequency

$\frac{1}{\sqrt{LC}}$ = natural frequency

Graph 1: "X Vs ω"

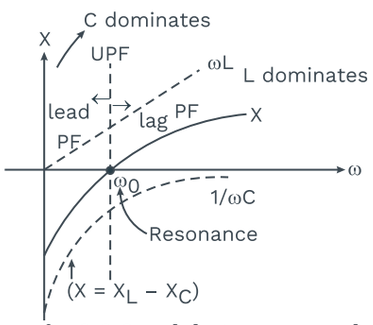


Fig. 8.3 Graph between X and ω

$$\left[Z = R + j\left(\omega L - \frac{1}{\omega C}\right)\right]$$

So instead of drawing X_C, we have drawn $-X_C$.

Rack Your Brain

What are the applications of resonance?

Grey Matter Alert!

The quality factor of the resonant curve depends upon the numerical value of RLC components.

Graph 2: "Z Vs ω"

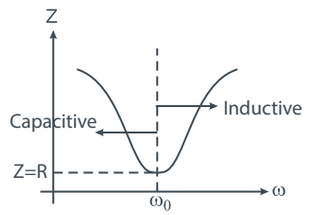


Fig. 8.4 Graph between Z and ω

$$Z = \left[R^2 + (X_L - X_C)^2\right]^{\frac{1}{2}}$$

- We are adding the square of two terms.
- Z = minimum if $X_L - X_C = 0$

[$\because (X_L - X_C)^2$ cannot be negative]

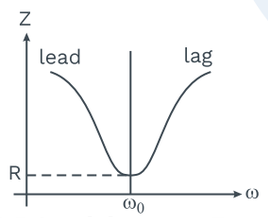


Fig. 8.5 Graph between Z and ω

At low frequency $\frac{1}{\omega C}$ = high.

So, capacitor dominates, and we get the leading power factor.

At high-frequency ωL = high.
So, the inductor dominates, and we get a lagging power factor.

Graph 3: "Y Vs ω"

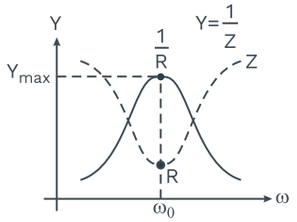
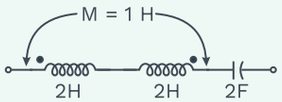


Fig. 8.6 Graph between Y and ω

At resonance, the network offers minimum opposition to the flow of current.

Previous Years' Question

The resonant frequency of the series circuit as shown in the figure is:



I) $\frac{1}{4\pi\sqrt{3}}$ Hz

II) $\frac{1}{4\pi}$ Hz

III) $\frac{1}{2\pi\sqrt{10}}$ Hz

IV) $\frac{1}{4\pi\sqrt{2}}$ Hz

Graph 4: "|I| Vs ω"

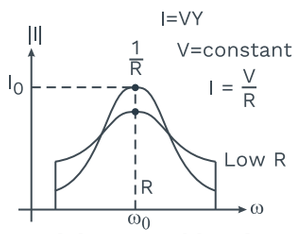


Fig. 8.7 Graph between |I| and ω

At resonance, Z = R

$$I = \frac{V}{R}$$

Graph 5: "ϕ Vs ω"

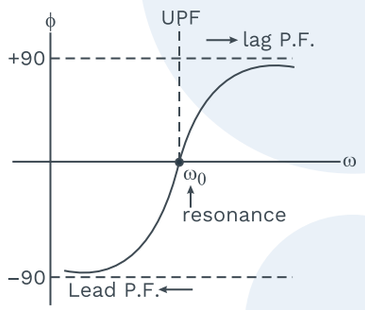


Fig. 8.8 Graph between ϕ and ω

$$\phi = \tan^{-1}\left(\frac{X_L - X_C}{R}\right) = \tan^{-1}\left[\frac{\omega L - \frac{1}{\omega C}}{R}\right]$$

At $\omega = 0$, $\phi = \tan^{-1}\left(\frac{-\infty}{R}\right) = -90°$

At $\omega \to \infty$, $\phi = \tan^{-1}\left(\frac{\infty}{R}\right) = 90°$

Important points:

1) At resonance, X = 0 and Z = minimum or Y = maximum.

2) I = VY. So, the current is also maximum at resonance.

3) At resonance power factor (cos θ) = 1 (unity)

4) Below resonance frequency, the network operates at the leading power factor and the above resonance frequency, the network operates at a lagging power factor.

Rack Your Brain

How do you calculate an RLC circuit's resonant frequency?

PHASOR DIAGRAMS

At resonance power factor,

At $\omega_0 = \omega_0$,

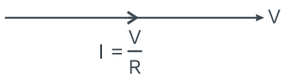


Fig. 8.9 Phasor diagram between V and R when $\omega = \omega_0$

At $\omega_0 < \omega_0$,

Capacitor dominates, so I leads V

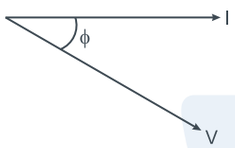


Fig. 8.10 Phasor diagram between V and R when $\omega < \omega_0$

$$\phi = \tan^{-1}\left(\frac{X_C - X_L}{R}\right)$$

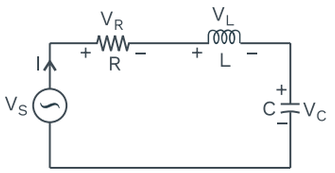


Fig. 8.11 Circuit diagram for series RLC circuit

At $\omega > \omega_0$,

Inductor dominates, so I lags V

$$\phi = \tan^{-1}\left(\frac{X_L - X_C}{R}\right)$$

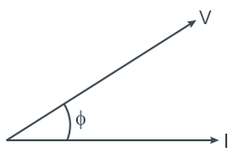


Fig. 8.12 Phasor diagram between V and R when $\omega > \omega_0$

Resonance $X_L = X_C \Rightarrow V_L = V_C$

$V_S = V_R$

Previous Years' Question

In a series RLC high quality factor (Q) circuit, the current peaks at a frequency:

I) Equal to the resonant frequency

II) Greater than the resonant frequency

III) Less than the resonant frequency

IV) None of the above is true

Grey Matter Alert!

For a tuned network, the quality factor needs to be high, the bandwidth should be small and therefore resistance used in the network should be small.

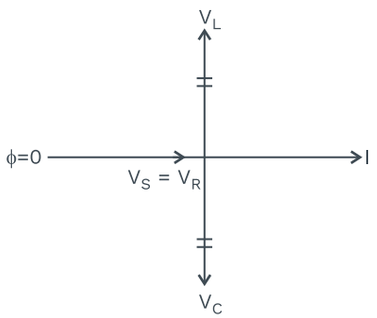


Fig. 8.13 Graph for V_L vs I

Variation of voltage across passive elements with change in frequency.

- At resonance $V_L = V_C$,
  I = maximum
  $\therefore$ V_R = IR is also maximum
- V_C becomes maximum at a frequency below the resonance frequency
- V_L = maximum at $\omega > \omega_0$.

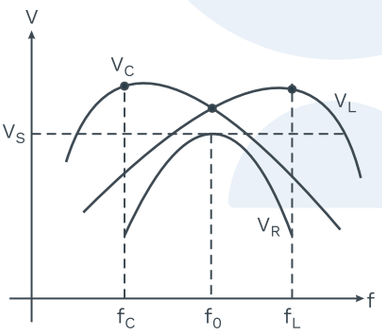


Fig. 8.14 Graph for V vs f

$$Z = R + j\left(\omega L - \frac{1}{\omega C}\right)$$

$$|Z| = \left[R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2\right]^{\frac{1}{2}}$$

$$|I| = \frac{V}{|Z|} = \frac{V}{\left[R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2\right]^{\frac{1}{2}}}$$

$$V_C = |I| X_C = \frac{V}{\omega C\left[R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2\right]^{\frac{1}{2}}}$$

Rack Your Brain

What are the advantages of resonance?

For maximum V_C, denominator = minimum,

$$\frac{d}{d\omega}\left[\omega^2C^2\left[R^2+\left(\omega L-\frac{1}{\omega C}\right)^2\right]\right]=0$$

$$2\omega C^2R^2+\frac{d}{d\omega}\left[\omega^2LC-1\right]^2=0$$

$$2\omega C\left[R^2C+2\omega^2L^2C-2L\right]=0$$

$$\omega = 0$$

$$2\omega^2L^2C = 2L - R^2C$$

$$\omega^2=\frac{1}{LC}-\frac{R^2}{2L^2}$$

$$\omega=\left[\frac{1}{LC}-\frac{R^2}{2L^2}\right]^{\frac{1}{2}}<\frac{1}{\sqrt{LC}}$$

For inductor voltage:

$$V_L = |I|X_L$$

$$=\frac{V.\omega L}{\left[R^2+\left(\omega L-\frac{1}{\omega C}\right)^2\right]^{\frac{1}{2}}}$$

$$=\frac{V}{\left[\frac{R^2}{\omega^2L^2}+\left(1-\frac{1}{\omega^2LC}\right)^2\right]^{\frac{1}{2}}}$$

For maximum V_L, denominator = minimum,

$$\frac{d}{d\omega}\left[\frac{R^2}{\omega^2L^2}+\left(1-\frac{1}{\omega^2LC}\right)^2\right]=0$$

$$\frac{-2R^2}{\omega^3L^2}+2\left(1-\frac{1}{\omega^2LC}\right)\times\frac{2}{\omega^3LC}=0$$

$$\frac{2}{\omega^3L}\left[\frac{2}{C}\left(1-\frac{1}{\omega^2LC}\right)-\frac{R^2}{L}\right]=0$$

$$\omega\to\infty$$

$$1-\frac{1}{\omega^2LC}=\frac{R^2C}{2L}$$

Previous Years' Question

For the series R-L-C circuit of figure-1, the partial phasor diagram at a certain frequency is shown in figure-2. The operating frequency of the circuit is:

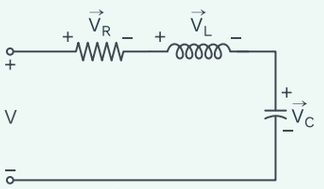


Figure-1

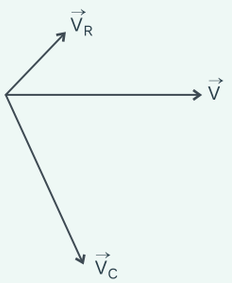


Figure-2

I) Equal to the resonance frequency
II) Less than the resonance frequency
III) Greater than the resonance frequency
IV) Not zero

$$\frac{1}{\omega^2 LC} = 1 - \frac{R^2C}{2L}$$

$$\frac{1}{\omega^2} = LC - \frac{R^2C^2}{2}$$

$$\omega = \frac{1}{\left[LC - \frac{R^2C^2}{2}\right]^{\frac{1}{2}}} > \frac{1}{\sqrt{LC}}$$

V_C = maximum at $\omega < \omega_0$
V_L = maximum at $\omega > \omega_0$
V_R = maximum at $\omega = \omega_0$

BANDWIDTH (B.W.)

Bandwidth is defined as the range of frequencies for which the power level in a signal is atleast half of the maximum power.

Half power frequency:

$$\frac{P_0}{2} = \frac{I_{0(rms)}^2 R}{2} = \left(\frac{I_{0(rms)}}{\sqrt{2}}\right)^2 R$$

$\omega_1 \rightarrow$ Lower cut off/roll off/corner frequency
$\omega_2 \rightarrow$ Upper cut off/roll off/corner frequency
$\omega_2 - \omega_1 = \frac{R}{L}$ rad/sec $\rightarrow$ Bandwidth

$$f_2 - f_1 = \frac{R}{2\pi L} \text{Hz}$$

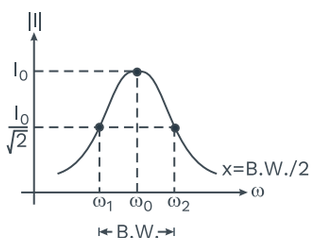


Fig. 8.15 Graph for I vs ω

- Half power frequency corresponds to ω at which $P = \frac{P_0}{2}$ and $I = \frac{I_{0(rms)}}{\sqrt{2}}$.
- This frequency is also called as cut-off frequency.
- In dB, power (dB) = 10 $\log_{10}$ P
  Current (dB) = 20 $\log_{10}$ I

At half power frequency $I = \frac{I_0}{\sqrt{2}}$

$$I(dB) = 20\log_{10}\frac{I_0}{\sqrt{2}}$$

$$= 20\log_{10} I_0 - 20\log\sqrt{2}$$

$$= 20\log_{10} I_0 - 3.0\,dB$$

$\therefore$ The cut-off frequency is also called as 3 dB frequency.

$$I = \frac{I_0}{\sqrt{2}}$$

$$\frac{V}{Z} = \frac{V/R}{\sqrt{2}}$$

$$Z = \sqrt{2}\,R \text{ at } \omega = \omega_C$$

$$Z^2 = 2R^2$$

$$R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2 = 2R^2$$

$$\left(\omega L - \frac{1}{\omega C}\right)^2 = R^2$$

$$\left(\omega L - \frac{1}{\omega C}\right) = \pm R$$

Case 1:

$$\omega^2 LC - 1 = \omega RC$$

$$\omega^2 LC - \omega RC - 1 = 0$$

$$\omega = \frac{RC + \sqrt{R^2C^2 + 4LC}}{2LC}$$

$$\omega_2 = \frac{R}{2L} + \sqrt{\frac{1}{LC} + \frac{R^2}{4L^2}}$$

Case 2:

$$\omega^2 LC - 1 = -\omega RC$$

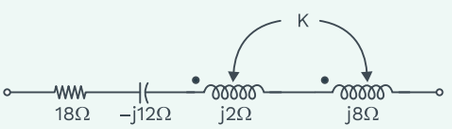
$$\omega^2 LC + \omega RC - 1 = 0$$

Grey Matter Alert!

A series RLC network represents a bandpass filter.

Previous Years' Question

In the series circuit shown in the figure, for series resonance, the value of the coupling coefficient K will be:



I) 0.25 **II)** 0.5
III) 0.999 **IV)** 1.0

$$\omega_1 = -\frac{R}{2L} + \left[\frac{1}{LC} + \frac{R^2}{4L^2}\right]^{\frac{1}{2}}$$

$$\omega_1.\omega_2 = \frac{1}{LC} + \frac{R^2}{4L^2} - \left(\frac{R}{2L}\right)^2 = \frac{1}{LC}$$

$$\omega_1\omega_2 = \omega_0^2$$

$$\omega_0 = \left[\omega_1\omega_2\right]^{\frac{1}{2}}$$

$$\log \omega_0 = \frac{\log \omega_1 + \log \omega_2}{2}$$

Rack Your Brain

Is resonance good or bad?

- Resonance frequency is the geometric mean of cut-off frequencies, but if the frequency is taken in the log, then it becomes arithmetic mean.

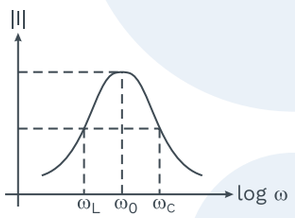


Fig. 8.16 Graph for I vs log ω

- The resonance frequency is the geometric mean of frequencies.

$$\omega_0 = \sqrt{\omega_1\omega_2} \text{ or } f_0 = \sqrt{f_1 f_2}$$

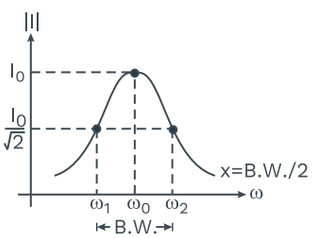


Fig. 8.17 Graph for I vs log ω

ω_0 is not the midpoint of ω_1 and ω_2, but log ω_0 is the midpoint.

Frequency selective circuit:

- The current or response of the circuit is high at some frequencies but low at other frequencies, so we say the circuit selects and passes only a few frequencies (lying in bandwidth) but stops other frequencies.

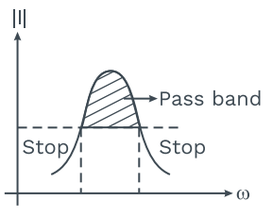


Fig. 8.18 Graph for |I| vs ω

- Series RLC circuit acts as bandpass filter.
- Bandwidth = $\frac{R}{L}$ rad/sec.

  Quality factor = $\omega_0 \frac{L}{R} = \frac{\omega_0}{B.W.}$

  $Q \propto \frac{1}{B.W.}$
- Bandwidth indicates the frequencies over which the response is spread.

  Radio Mirchi → 98.3 MHz (resonance)

At 98.4 MHz, we can hear radio but distorted which means signals spread over the range of frequencies.

- One signal must not interfere with other signals so,

  B.W. = low, Q = high
- Quality factor (Q) indicates the sharpness of the resonance curve and high quality factor (Q) indicates narrow bandwidth or frequency selective circuit.

Rack Your Brain

What are some real-life examples of resonance?

SELECTIVITY (S)

Selectivity is the ability of a circuit to distinguish between desired and undesired frequencies.

$$S \propto \frac{1}{B.W.}$$

$$S = \frac{f_0}{f_2 - f_1} = \frac{\frac{1}{2\pi\sqrt{LC}}}{\frac{R}{2\pi L}} = \frac{1}{R}\sqrt{\frac{L}{C}}$$

$$S = Q_0$$

Example 1:
Consider a series RLC circuit with a sinusoidal ac voltage as shown below.



Fig. 8.19

Determine:

a) The resonant frequency and the half-power frequencies.

b) The quality factor and the bandwidth.

c) The amplitude of the current at frequencies ω_0, ω_1 and ω_2.

Sol:

a) Given, R = 2Ω, L = 1 mH, C = 0.4 μF, V_m = 20V

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{1\times 10^{-3}\times 0.4\times 10^{-6}}} = 50\,\text{krad / sec}$$

The lower half-power frequency is:

$$\omega_1 = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

$$= -\frac{2}{2\times 10^{-3}} + \sqrt{\left(\frac{2}{2\times 10^{-3}}\right)^2 + \frac{1}{1\times 10^{-3}\times 0.4\times 10^{-6}}}$$

ω_1 = 49 krad/sec

The upper half-power frequency is:

$$\omega_2 = \frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

$$= \frac{2}{2\times 10^{-3}} + \sqrt{\left(\frac{2}{2\times 10^{-3}}\right)^2 + \frac{1}{1\times 10^{-3}\times 0.4\times 10^{-3}}}$$

= 51 krad/sec

b) $Q = \frac{\omega_0 L}{R} = \frac{50\times 10^3 \times 1\times 10^{-3}}{2} = 25$

$$B.W = \frac{R}{L} = \frac{2}{1\times 10^{-3}} = 2\,\text{krad / sec}$$

c) At $\omega = \omega_0$, $I = \frac{V_m}{R} = \frac{20}{2} = 10\,A$

At $\omega = \omega_1, \omega_2$, $I = \frac{V_m}{\sqrt{2}R} = \frac{20}{2\times\sqrt{2}} = 7.071A$

Example 2:

A series RLC circuit with R = 1Ω, L = 0.2 H, and an unknown C is excited with an a.c. source of 100 V, 50 Hz. For resonance condition, calculate-

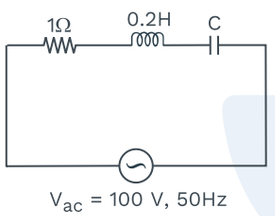


Fig. 8.20

a) the capacitance C,
b) the voltage across C, and
c) the Q-factor

Rack Your Brain

At what frequency do humans vibrate?

Sol:

Given,

R = 1Ω, L = 0.2 H, V_{ac} = 100V and f = 50 Hz

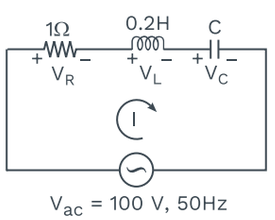


Fig. 8.21

$\therefore\ \omega = 2 \times \pi \times f = 2 \times \pi \times 50 = 314$ rad/sec

a) For resonance condition,

$$\omega L = \frac{1}{\omega C}$$

$$\omega^2 L = \frac{1}{C}$$

$$C = \frac{1}{\omega^2 L} = \frac{1}{(2\times\pi\times 50)^2 \times 0.2} = 50.712\,\mu F$$

b) The current in the circuit under resonance:

$$I = \frac{V}{Z} = \frac{V}{R} = 100\,A$$

$$V_C = \frac{1}{C}\int I\,dt = \frac{1}{C} \times \frac{I}{j\omega} = \frac{I}{j\omega C} = \frac{I}{\omega C} \angle -90°$$

$$V_C = \frac{100\angle -90°}{2 \times \pi \times 50 \times 50.712 \times 10^{-6}} = 6279.99\angle -90°$$

$$|V_C| = 6279.99\,V$$

c) The quality factor for series resonance circuit:

$$Q = \frac{1}{R}\sqrt{\frac{L}{C}} = \frac{\omega L}{R} = \frac{1}{\omega RC} = \frac{1}{1}\sqrt{\frac{0.2}{50.712 \times 10^{-6}}}$$

Q = 62.8

Series resonant circuit works as a voltage magnification/amplification circuit.

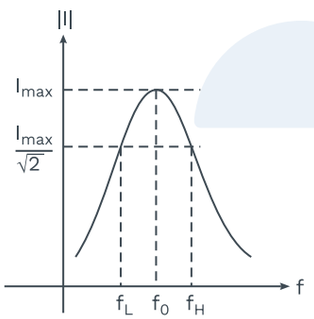


Fig. 8.22

Example 3:

A series RLC circuit has a resonance frequency of 50 rad/sec and a quality factor of 80. If the impedance at resonance is 10Ω. Find the value of R_L and C.

Sol:

Given,

Q = 80, Z = 10Ω,

We know, at resonance

Z = R = 10Ω

Quality factor, $Q = \frac{\omega_0 L}{R} = 80$

$$L = \frac{R(80)}{50}$$

$$L = \frac{10 \times 80}{50} \qquad \{\omega_0 = 50\,\text{rad/sec}\}$$

$L = 16$ H

Resonant frequency,

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

$$C = \frac{1}{\omega_0^2 L}$$

$$C = \frac{1}{(50)^2 (16)}$$

$C = 25\ \mu F$

Example 4:

The resonance frequency of the given circuit is:

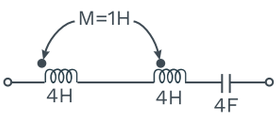


Fig. 8.23

Sol:

We know,

Resonance frequency,

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

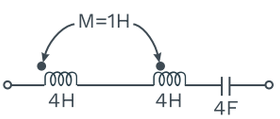


Fig. 8.24

Given circuit is in additive polarity, so:

$L_{eq} = L_1 + L_2 + 2M$

$L_{eq} = 4 + 4 + 2 \times 1 = 10H$

$$f_0 = \frac{1}{2\pi\sqrt{10 \times 4}} = \frac{1}{2\pi\sqrt{40}}$$

$$f_0 = \frac{1}{4\pi\sqrt{10}} \text{Hz}$$

Example 5:
A series RLC circuit has R = 50Ω, L = 100 μH and C = 1 μF. Find the lower half power frequency of the circuit.

Sol:
Given, R = 50Ω, L = 100 μH and C = 1 μF
Resonant frequency,

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

$$\omega_0 = \frac{1}{\sqrt{100 \times 10^{-6} \times 1 \times 10^{-6}}}$$

$$\omega_0 = \frac{1}{10 \times 10^{-6}}$$

$$\omega_0 = 10^5 \text{ rad/sec}$$

Quality factor, $Q = \frac{\omega_0 L}{R}$

$$Q = \frac{10^5 \times 100 \times 10^{-6}}{50}$$

$$Q = 0.2$$

Lower half-power frequency,

$$\omega_1 = \omega_0 \left[\sqrt{1 + \left(\frac{1}{2Q}\right)^2} - \frac{1}{2Q}\right]$$

$$\omega_1 = 10^5 \left[\sqrt{1 + \left(\frac{1}{2 \times 0.2}\right)^2} - \frac{1}{2 \times 0.2}\right]$$

$$\omega_1 = 10^5 \left[\sqrt{7.25} - 2.5\right]$$

$$\omega_1 = 0.192 \times 10^5 = 19.2 \text{ kHz}$$

Previous Years' Question

A DC voltage source is connected across a series R-L-C circuit. Under steady-state conditions, the applied DC voltage drops entirely across the:

I) R only
II) L only
III) C only
IV) R and L combination

Example 6:

A series RLC circuit is driven by a variable frequency voltage, as shown in the figure. Then obtain the average power dissipated by the network at resonance frequency.

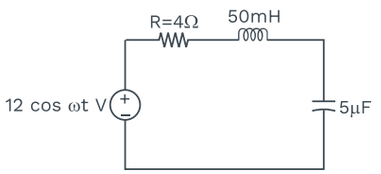


Fig. 8.25

Sol:

Given circuit,

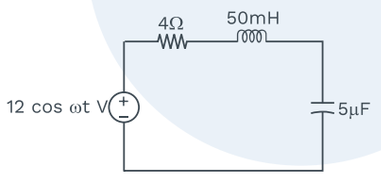


Fig. 8.26

At resonance,

$X_L = X_C$

Then the circuit becomes a resistive circuit,

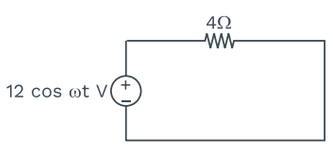


Fig. 8.27

Average power $= \frac{\left(\frac{V_m}{\sqrt{2}}\right)^2}{R} = \frac{\left(\frac{12}{\sqrt{2}}\right)^2}{4} = \frac{144}{2\times 4} = 18$ watts

Example 7:

Find R so that the circuit shown in the figure resonates at supply frequency.

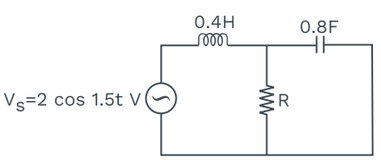


Fig. 8.28

Sol:

The equivalent impedance of the circuit from the source side is,

$$Z_{eq} = j\omega L + \frac{R \times \frac{1}{j\omega C}}{R + \frac{1}{j\omega C}}$$

$$= j\omega L + \frac{R}{1 + j\omega RC}$$

$$= j(1.5 \times 0.4) + \frac{R}{1 + j(1.5 \times R \times 0.8)}$$

$$= \frac{R}{1 + 1.44R^2} + j\left(0.6 - \frac{1.2R^2}{1 + 1.44R^2}\right)$$

At resonance, the imaginary part becomes zero.

$$0.6 - \frac{1.2R^2}{1 + 1.44R^2} = 0$$

$$\therefore \quad R = \sqrt{\frac{1}{1.2 \times 0.47}} = 1.33\Omega$$

Example 8:

Impedances Z_2 and Z_3 in parallel are in series with an impedance Z_1 across a 100 V, 50 Hz ac supply. Z_1 = 6.25 + j1.25 Ω, Z_2 = (5 + j0) and Z_3 = (5 – jX_C) Ω. Determine the value of capacitance of X_C such that the total current of the circuit will be in phase with the supply voltage. Find the current and power in the circuit?

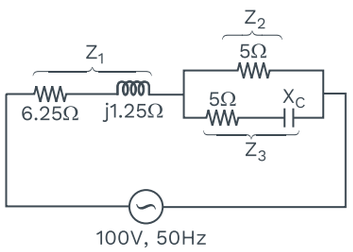


Fig. 8.29

Sol:

Total impedance, $Z_T = Z_1 + \frac{Z_2 Z_3}{Z_2 + Z_3}$

$$= 6.25 + j1.25 + \frac{5(5 - jX_C)}{5 + (5 - jX_C)}$$

$$= 6.25 + j1.25 + \left(\frac{25 - j5X_C}{10 - jX_C}\right) \times \left(\frac{10 + jX_C}{10 + jX_C}\right)$$

$$= 6.25 + j1.25 + \frac{\left(250 + 5X_C^2\right) - j25X_C}{100 + X_C^2}$$

$$= 6.25 + \frac{250 + 5X_C^2}{100 + X_C^2} - j\left(\frac{25X_C}{100 + X_C^2} - 1.25\right)$$

If the total current in the circuit is in phase with the applied voltage, Z_r will be purely resistive. i.e.

$$\frac{25X_C}{100 + X_C^2} - 1.25 = 0 \quad \text{or } X_C = 10$$

(where Z_r = impedance at resonance)

$$\text{or } \frac{1}{\omega C} = 10$$

$$\therefore \quad C = \frac{1}{10 \times 2 \times 3.14 \times 50} = 318\,\mu F$$

Putting $X_C = 10\ \Omega$ in the expression of Z_r, we obtain $Z_r = 10\ \Omega$.

The circuit current, $I = \frac{100}{10} = 10A$

and the total power of the circuit = $I^2Z_r = 10^2 \times 10 = 1000W$

Rack Your Brain

Is a higher natural frequency better?

Example 9:

A series RLC circuit consists of R = 25Ω, L = 100 μH and C = 1000 pF which resonates at f_0 = 500 kHz. Obtain-

a) The quality factor of the circuit

b) The new value of capacitance which resonates at 500 kHz, if the value of inductance is doubled, then find the value of the quality factor.

Sol:

a) $Q_1 = \frac{1}{R}\sqrt{\frac{L}{C}} = \frac{1}{25}\sqrt{\frac{100 \times 10^{-6}}{1000 \times 10^{-12}}} = 12.6$

b) The new value of L is 200 μH

Now, $f_0 = \frac{1}{2\pi\sqrt{LC}}$

$$f_0^2 = \frac{1}{4\pi^2 LC}$$

$$C = \frac{1}{4\pi^2 L f_0^2}$$

$$C = \frac{1}{4\pi^2 \times \left(500 \times 10^3\right)^2 \times 200 \times 10^{-6}}$$

$$C = 500 \times 10^{-12}\ F = 500\ pF$$

$$Q_2 = \frac{1}{R}\sqrt{\frac{L}{C}} = \frac{1}{25}\sqrt{\frac{200 \times 10^{-6}}{500 \times 10^{-12}}} = 25$$

Example 10:
A coil of resistance 100Ω and inductance of 100 μH is connected in series with a 100 pF capacitor. The circuit is connected to a 10 V variable frequency source. Calculate:
a) the resonant frequency
b) current of resonance
c) the voltage across L and C at resonance
d) Q-factor of the circuit.

Sol:
Given,
Resistance, R=100Ω
Inductance, L=100 μH = 100×10^{-6} H
Capacitance, C=100 pF = 100×10^{-12} H
a) Resonant Frequency,

$$f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{100 \times 10^{-6} \times 100 \times 10^{-12}}}$$

$$f_0 = 1.59 \times 10^6 Hz$$

$$f_0 = 1.59\ MHz$$

b) Current at resonance, $I_r = \frac{V}{R} = \frac{10}{100} = 0.1\ A$

c) At resonance,

$$X_L = 2\pi f_r L = 2\pi \times 1.59 \times 10^6 \times 100 \times 10^{-6}$$

$$X_L = 1000\ \Omega$$

At resonance,
Voltage across the inductor, $V_L = I_r X_L = 0.1 \times 1000 = 100\ V$
Voltage across the capacitor, $V_C = I_r X_C = 0.1 \times 1000 = 100\ V$

Grey Matter Alert!

For a parallel RLC circuit, the quality factor of the network represents the current amplification factor and must have a high value for any tuned circuit.

d) Q-factor $= \frac{1}{R}\sqrt{\frac{L}{C}}$

$$Q = \frac{1}{100}\sqrt{\frac{10^{-4}}{10^{-10}}} = 10$$

Note: At resonance, the voltage across L or C is Q times the applied voltage.

Example 11:
A series R-L-C circuit has Resistance, R= 5Ω., inductance, L=0.2 H, and capacitance C = 50μF. The applied voltage is 200 V. Determine:

a) Resonant frequency
b) Quality factor (Q factor)
c) Bandwidth
d) Upper and lower half-power frequencies
e) Current at resonance
f) Current of half-power points
g) Voltage across inductance at resonance.

Sol:
Given,

$R = 5\Omega$

$L = 0.2H$

$C = 50 \times 10^{-6}$ F

$V = 200$ volts

a) Resonant frequency,

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

$$\frac{1}{2\pi\sqrt{0.2 \times 50 \times 10^{-6}}}$$

$f_0 = 50.33$ Hz

b) Quality factor,

$$Q = \frac{\omega_0 L}{R} = \frac{2\pi f_0 L}{R}$$

Previous Years' Question

The current i(t), through a 10-Ω resistor in series with an inductance, is given by
i(t) = 3 + 4 sin (100 t + 45°) + 4 sin (300 t + 60°) Amperes.
The RMS value of the current and the power dissipated in the circuit are:

I) $\sqrt{41}$ A , 410 W, respectively
II) $\sqrt{35}$ A , 350 W, respectively
III) 5 A, 250 W, respectively
IV) 11 A, 1210 W, respectively

$$Q = \frac{2\pi \times 50.33 \times 0.2}{5}$$

$$Q = 12.65$$

c) Bandwidth, $BW = \frac{\omega_0}{Q} = \frac{2\pi \times 50.33}{12.65} = 3.98$ Hz

d) Upper half-power frequency,

$$f_1 = f_0 + \frac{B.W}{2}$$

$$= 50.33 + \frac{3.98}{2}$$

$$= 52.32 \text{ Hz}$$

e) Lower half-power frequency,

$$f_2 = f_0 - \frac{B.W}{2}$$

$$= 50.33 - \frac{3.98}{2}$$

$$= 48.34 \text{ Hz}$$

f) Current at resonance,

$$I_r = \frac{V}{R} = \frac{200}{5} = 40 \text{ A}$$

g) Current at half-power points
$= 0.707 I_r$
$= 0.707 \times 40$
$= 28.28$ A

h) Voltage across L at resonance
$= I_r X_L = I_r \times 2\pi f_0 L$
$= 40 \times 2\pi \times 50.33 \times 0.2$
$= 2529.87$ V

Grey Matter Alert!

A parallel RLC network behaves as a band stop filter or band-reject filter.

PARALLEL RESONANCE

General parallel resonant circuit consisting of ideal elements R, L, and C are shown in the figure. The circuit is said to be in resonance when the resultant susceptance of the circuit is zero.

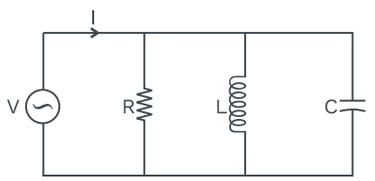


Fig. 8.30

Admittance $Y = \frac{1}{R} + j\omega C - \frac{j}{\omega L}$

$$Y = \frac{1}{R} + j\left(\omega C - \frac{1}{\omega L}\right)$$

For resonance,
Imaginary(Y) = 0

$$\omega C - \frac{1}{\omega L} = 0 \text{ or } \omega = \frac{1}{\sqrt{LC}},$$

$$f = \frac{1}{2\pi\sqrt{LC}} \rightarrow \text{resonance frequency}$$

- Resonance occurs at $\omega = \frac{1}{\sqrt{LC}}$ rad / sec
- At resonance $Y = \frac{1}{R}$ or Z = R.
- At any other frequency

$$Y = \left[\left(\frac{1}{R}\right)^2 + \left(\omega C - \frac{1}{\omega L}\right)^2\right]^{\frac{1}{2}} > \frac{1}{R}.$$

We can understand admittance is minimum at resonance.
∴ In a parallel resonance circuit, the flow of current is minimum.

Previous Years' Question

A series R-L-C circuit has a Q of 100 and an impedance of (100 + j0) Ω at its resonant angular frequency of 10^7 radians/sec. The values of R and L are:
R = _____ ohms.
L = _____ Henries.

Graph 1: "B Vs ω"

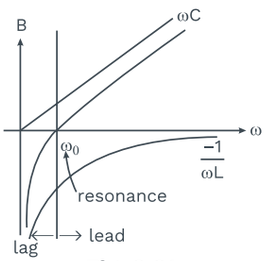


Fig. 8.31

$$\left[Y = \frac{1}{R} + j\left(\omega C - \frac{1}{\omega L} \right) \right]$$

Rack Your Brain

What is Earth's network frequency?

Graph 2: "Y Vs ω"

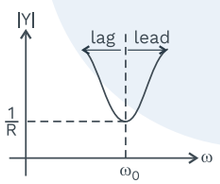


Fig. 8.32

Admittance is minimum at resonance.

Graph 3: "Z Vs ω"

I = constant

V = IZ

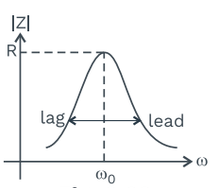


Fig. 8.33

Graph 4: "V Vs ω"

I = constant

V = IZ

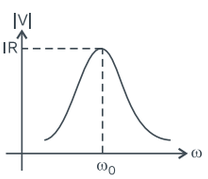


Fig. 8.34

Graph 5: "ϕ Vs ω"

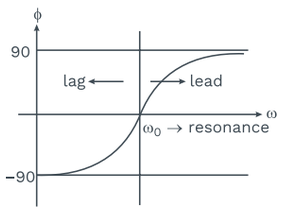


Fig. 8.35

$$Y = \frac{1}{R} + j\left(\omega C - \frac{1}{\omega L}\right)$$

$$\phi = \tan^{-1}\left[\frac{\omega C - \frac{1}{\omega L}}{\frac{1}{R}}\right]$$

at $\omega \to 0$, $\phi = -90°$

at $\omega \to \infty$, $\phi = 90°$

Phasor diagrams:

At $\omega = \omega_0$,

Y = G (purely resistive)

"I" in phase with V (unity power factor)



Fig. 8.36

I → Current

V → Voltage

V = IR

At $\omega < \omega_0$,

$Y = G - j\, B_{net}$ (R-L circuit)

"I" lag V by $\phi < 90°$ (lagging)

I → Current

V → Voltage

Inductor dominates capacitor $\left(\frac{1}{\omega L} > \omega C\right)$

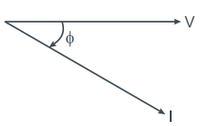


Fig. 8.37

Rack Your Brain

What hertz of frequency is harmful?

$$\phi = \tan^{-1}\left(\frac{\frac{1}{\omega L} - \omega C}{\frac{1}{R}}\right)$$

At $\omega > \omega_0$,

$Y = G + j\, B_{net}$ (R-C circuit)

"I" leads V by $\phi < 90°$ (leading)

$I \rightarrow$ Current

$V \rightarrow$ Voltage

Capacitor dominates inductor $\left(\omega C > \frac{1}{\omega L}\right)$

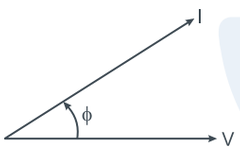


Fig. 8.38

$$\phi = \tan^{-1}\left(\frac{\omega C - \frac{1}{\omega L}}{\frac{1}{R}}\right)$$

$$Y = \frac{1}{R} + j\left(\omega C - \frac{1}{\omega L}\right)$$

At resonance:

$|B_L| = |B_C| \Rightarrow |I_L| = |I_C|$

$I = I_R$

$\phi = 0$, $\cos\phi = 1$

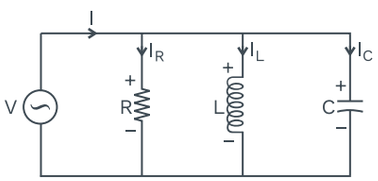


Fig. 8.39

$$I = \frac{V}{X} = VB$$

Previous Years' Question

In the circuit shown, at resonance, the amplitude of the sinusoidal voltage (in Volts) across the capacitor is:



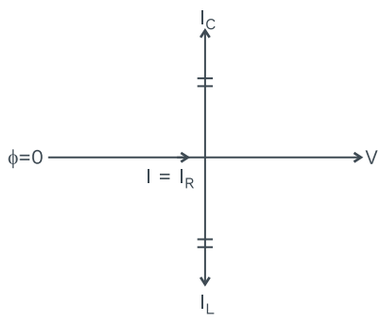


Fig. 8.40

$$I_R = \frac{V}{R}$$

Q factor at resonance:

$$Q_0 = \frac{R}{\omega_0 L} = \omega_0 RC$$

But, $\omega_0 = \frac{1}{\sqrt{LC}}$

$$Q_0 = \frac{R}{\frac{1}{\sqrt{LC}} \times L} = R\sqrt{\frac{C}{L}}$$

* Reciprocal of series RLC.

$$Q = \frac{\text{reactive power}}{\text{real power}} = \frac{V^2 / X_L}{V^2 / R} = \frac{V^2 / X_C}{V^2 / R}$$

$$= \frac{R}{X_L} = \frac{R}{\omega_0 L}$$

$$= \frac{R}{X_C} = \omega_0 RC$$

- In parallel resonance, the net impedance is maximum; therefore, current is minimum. So it is called a rejector circuit.
- In parallel resonance, the quality factor of the network represents the current amplification factor and must have a high value for any tuned circuit.
- At resonance, Z = maximum, I = minimum. A parallel RLC network behaves as a band-stop filter or bend-reject filter.
- At resonance, V = IR; I = Supply current

$$I_L = \frac{V}{X_L} = \frac{IR}{X_L} = I\left(\frac{R}{X_L}\right) = QI$$

$$I_C = \frac{V}{X_C} = \frac{IR}{X_C} = QI$$

- Current through inductor and capacitor is amplified by a factor Q concerning supply current at resonance.

Half power frequency:

$$Y(\omega_0) = \frac{1}{R}$$

At cut off or half power frequency.

$Y = \sqrt{2} / R$

$$\frac{\sqrt{2}}{R} = \left[\frac{1}{R^2} + \left(\omega C - \frac{1}{\omega L}\right)^2\right]^{\frac{1}{2}}$$

$$\frac{2}{R^2} = \frac{1}{R^2} + \left(\omega C - \frac{1}{\omega L}\right)^2$$

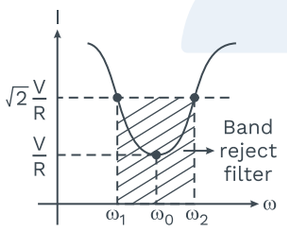


Fig. 8.41

If V = constant, then I follow the same curve as Y.

$$\frac{1}{R^2} = \left(\omega C - \frac{1}{\omega L}\right)^2$$

$$\omega C - \frac{1}{\omega L} = \pm\frac{1}{R}$$

$$\omega^2 LC - 1 = \pm\frac{\omega L}{R}$$

Case 1:

$$\omega^2 - \frac{\omega}{RC} - \frac{1}{LC} = 0$$

$$\omega_2 = \frac{1}{2RC} + \sqrt{\frac{1}{(2RC)^2} + \frac{1}{LC}}$$

Case 2:

$$\omega^2 + \frac{\omega}{RC} - \frac{1}{LC} = 0$$

$$\omega_1 = -\frac{1}{2RC} + \sqrt{\left(\frac{1}{2RC}\right)^2 + \frac{1}{LC}}$$

BANDWIDTH (B.W.)

Half power frequency:

$$\frac{P_0}{2} = \frac{V_{rms}^2}{2R} = \frac{1}{2}\left(\frac{V_m}{\sqrt{2}}\right)^2 \times \frac{1}{R}$$

ω_1 = Lower cut off/roll off/corner frequency
ω_2 = Upper cut off/roll off/corner frequency

$$\omega_2 - \omega_1 = \frac{1}{RC}\text{rad / sec} \leftarrow \text{Bandwidth}$$

$$f_2 - f_1 = \frac{1}{2\pi RC}\text{Hz}$$

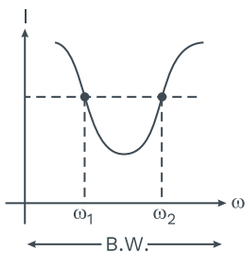


Fig. 8.42

- As resistance increases, bandwidth decreases.
  So, the curve becomes sharper.
- The resonance frequency is the geometric mean of bandwidth frequencies.

$$\omega_0 = \sqrt{\omega_1\omega_2} \text{ or } f_0 = \sqrt{f_1 f_2}$$

From graph-

$$\omega_0 - \omega_1 = \frac{\text{B.W.}}{2} \text{ and } \omega_2 - \omega_0 = \frac{\text{B.W.}}{2}$$

Previous Years' Question

The damping ratio of a series RLC circuit can be expressed as:

I) $\frac{R^2C}{2L}$ **II)** $\frac{2L}{R^2C}$

III) $\frac{R}{2}\sqrt{\frac{C}{L}}$ **IV)** $\frac{2}{R}\sqrt{\frac{L}{C}}$

- ω_0 is the midpoint of ω_1 and ω_2, which is not always true:

$$f_1 = f_0 - \frac{1}{4\pi RC}\text{Hz} \text{ and } f_2 = f_0 + \frac{1}{4\pi RC}\text{Hz}$$

$$Q = \frac{\omega_0}{\text{B.W.}} = R\sqrt{\frac{C}{L}}$$

This approximation works if BW <<< ω_0.

For problem solving:

$$\omega_1\omega_2 = \omega_0^2$$

$$\omega_2 - \omega_1 = \text{B.W.}$$

Solve these 2 equations to determine ω_1 and ω_2.

PRACTICAL PARALLEL RESONANCE (TANK CIRCUIT)

- A practical resonant parallel circuit contains an inductive coil of resistance R and inductance L in parallel with a capacitor C.
- It is called a tank circuit because it stores energy in the magnetic field of the coil and the electric field of the capacitor.
- Note that the resistance of the capacitor R_C is negligibly small.

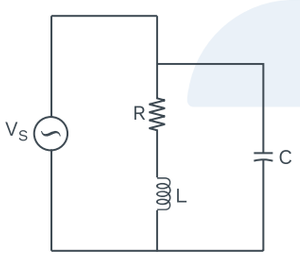


Fig. 8.43

$$Y = \frac{1}{R + j\omega L} + j\omega C = \frac{R - j\omega L}{R^2 + \omega^2 L^2} + j\omega C$$

For resonance imaginary(Y) = 0

$$\frac{\omega L}{R^2 + \omega^2 L^2} = \omega C$$

$$\frac{L}{C} - R^2 = \omega^2 L^2$$

$$\omega = \left[\frac{1}{LC} - \frac{R^2}{L^2}\right]^{\frac{1}{2}}$$

At resonance Y,

$$\text{real part} = \frac{R}{R^2 + \omega^2 L^2} = \frac{R}{\frac{L}{C}} = \frac{RC}{L}$$

Ideal tank (R = 0):

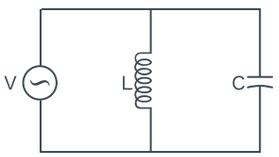


Fig. 8.44

$$Y = j\omega C - \frac{j}{\omega L}$$

At resonance Imaginary(Y) = 0

Y = 0 and Z = ∞

∴ Tank circuit acts as open circuit at resonance.

Dynamic impedance/resistance:

It is the resistance offered by the circuit to the input at the resonance condition.

a) For general series RLC:

$$Z_{dyn} = R$$

b) For general parallel RLC:

$$Z_{dyn} = R$$

c) For practical tank

$$Z_{dyn} = \frac{L}{RC}, V_{dyn} = \frac{R}{R^2 + X_L^2} = RC, Y = \frac{RC}{L}$$

For the ideal tank:

$$Z_{dyn} = \infty \text{ (open circuit)}$$

Important formula

Series RLC:

$$\omega_0 = \frac{1}{\sqrt{LC}};\ \text{B.W.} = \frac{R}{L};\ Q = \frac{\omega_0}{BW} = \frac{1}{R}\sqrt{\frac{L}{C}}$$

Frequency selective circuit (Band pass filter)

Parallel RLC:

$$\omega_0 = \frac{1}{\sqrt{LC}};\ \text{B.W.} = \frac{1}{RC};\ Q = \frac{\omega_0}{\text{B.W.}} = R\sqrt{\frac{C}{L}}$$

Frequency rejecting circuit (Band stop filter)

- $\omega_1 \omega_2 = \omega_0^2$; $\omega_2 - \omega_1 = \text{B.W.}$

Example 12:

Consider a parallel RLC circuit that consists of R = 100 kΩ, L = 20 mH and C = 5 nF.

Find:

a) ω_0

b) Quality factor (Q).

Sol:

a) $\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{20 \times 5 \times 10^{-12}}}$

$= \frac{10^6}{10} = 10^5$ rad / sec

= 100 K rad/sec

b) For parallel RLC circuit,
Q-factor -

$Q_0 = R\sqrt{\frac{C}{L}}$

$Q_0 = 100 \times 10^3 \sqrt{\frac{5 \times 10^{-9}}{20 \times 10^{-3}}}$

$Q_0 = 50$

Example 13:

Determine the resonant frequency (ω_0) in the given circuit.

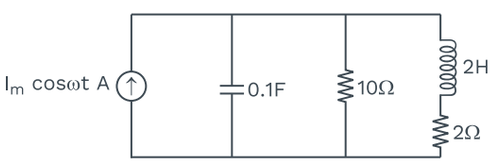


Fig. 8.45

Sol:

$$Y = j\omega(0.1) + \frac{1}{10} + \frac{1}{2 + j2\omega}$$

$$= j\omega(0.1) + 0.1 + \frac{2 - j2\omega}{4 + 4\omega^2}$$

Imaginary(Y) = 0

$$0.1\omega - \frac{2\omega}{4 + 4\omega^2} = 0$$

$$0.1\omega = \frac{2\omega}{4 + 4\omega^2}$$

$$20 = 4 + 4\omega^2$$

$$\omega = 2 \text{ rad/sec}$$

Example 14:

Design a parallel resonant RLC circuit with $\omega = 10$ rad/s, and Q = 20. Calculate the bandwidth of the circuit. Let R = 10Ω.

Sol:

$$\frac{1}{\sqrt{LC}} = 10$$

$$R\sqrt{\frac{C}{L}} = 20$$

$$BW = \frac{\omega_0}{Q} = 0.5 = \frac{1}{RC}$$

$$C = \frac{1}{5} = 0.2F$$

$$L = \frac{1}{100C} = 0.05H$$

Example 15:

The lower half frequency and the bandwidth of a parallel RLC circuit is 1000 rad/s and 100 rad/s respectively. If the value of resistance is 200Ω, then determine the value of L and C.

Sol:

$\omega_1 = 1000$ rad/sec

$\omega_2 = 1100$ rad/sec

$$BW = 100 = \frac{1}{200 \times C}$$

$C = 50\mu F$

$$\omega_0^2 = \omega_1\omega_2 = 11 \times 10^5 = \frac{1}{LC}$$

$L = 18.18$ mH

Example 16:
In the circuit shown below, find out the voltage across the resistor at the resonance frequency.

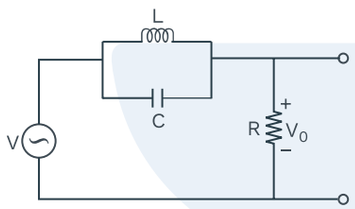


Fig. 8.46

Sol:
Given,

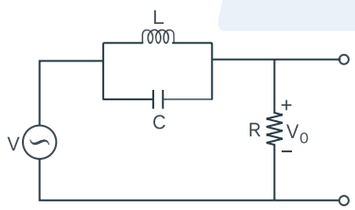


Fig. 8.47

At parallel resonance,

$X_L = X_C$

LC $\rightarrow$ open

Current, I = 0 A

then, voltage V_0 = 0 V

Chapter Summary



1) **Resonance:** Resonance is the condition at which the voltage across a circuit becomes in phase with the current supplied to the circuit.
   - At resonance, the circuit behaves like a resistive circuit.
   - The power factor of the circuit at resonance is unity.
2) **Resonant frequency:** The frequency at which impedance of the circuit is purely resistive is called resonant frequency.

$$\omega_0 = \frac{1}{\sqrt{LC}} \text{ rad/sec}$$

3) The half-power frequencies (ω_1, ω_2) are those frequencies at which the power dissipated is half of the maximum power dissipated. The geometric mean between the half-power frequencies is the resonant frequency.

$$\omega_0 = \sqrt{\omega_1 \omega_2}$$

4) The bandwidth is the frequency band between half-power frequencies.

$$\text{B.W.} = \omega_2 - \omega_1$$

Where ω_1 = lower half-power frequency
ω_2 = upper half-power frequency

5) The quality factor is a measure of the sharpness of the resonance peak. It is the ratio of resonant (angular) frequency to the bandwidth,

$$Q = \frac{\omega_0}{\text{B.W.}}$$

6) $$Q - \text{factor} = 2\pi\left(\frac{\text{Maximum energy stored}}{\text{Energy dissipated per cycle}}\right)$$

$$Q_0 = \frac{\omega_0}{\Delta\omega} = \frac{1}{\omega_0 RC} = \frac{\omega_0 L}{R} = \frac{1}{R}\sqrt{\frac{L}{C}}$$

(for series resonance)

7) Frequency at which voltage across the capacitor is maximum.

$$f_c = \frac{1}{2\pi}\sqrt{\frac{1}{LC} - \frac{R^2}{2L^2}}$$

(for series resonance)

8) Frequency at which voltage across the inductor is maximum,

$$f_L = \frac{1}{2\pi}\sqrt{LC - \frac{C^2R^2}{2}}$$

(for series resonance)

9) Selectivity,

$$\text{Selectivity} = \frac{\text{Resonance frequency}}{\text{Bandwidth}} = \frac{f_0}{f_2 - f_1}$$

10) Feature of series resonance circuit-

- The resonant frequency is the geometric mean of the two half-power frequencies.
- The quality factor represents the voltage amplification factor and must have a high value for any tuned circuit.
- For a tuned network, the quality factor must be high, the bandwidth should be small, and therefore, the resistance used in the network should be small.
- Any series RLC network represents a bandpass filter.

11) For parallel RLC circuit:

- **Bandwidth:** $\Delta\omega = (\omega_2 - \omega_1) = \frac{1}{RC}$
- **Q-factor:** $Q_0 = \frac{\omega_0}{\Delta\omega} = \omega_0 RC = \frac{R}{\omega_0 L} = R\sqrt{\frac{C}{L}}$
- The quality factor of the network represents the current amplification factor and must have a high value for any tuned circuit.
- For a tuned network, the quality factor should be high, and therefore should have low bandwidth, and the capacitance value should be high.
- A parallel RLC network behaves as a band stop filter or band-reject filter.

PRACTICE QUESTIONS

Q1 A series RLC circuit is excited by a voltage e = E sin ωt, where $LC < \frac{1}{\omega^2}$. Which of the given statement(s) is/are correct?

a) Current lags the applied voltage
b) Current leads the applied voltage
c) Current is in phase with the applied voltage
d) Voltage across L and C are equal

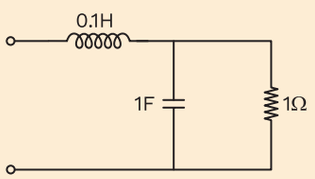
Q2 A coil has a resistance of 5 ohms and inductance of 0.1 H connected in series with a condenser of capacitance of 50 μF. A constant alternating voltage of 200 volts is applied to the circuit. The voltage across the inductor at resonance is:

a) 200 volts b) 1800 volts c) 1788 volts d) 2000 volts

Q3 For a parallel RLC circuit $\omega_0 = 10^8$ and Q = 20. Given C = 20 pF. Find the value of R at resonance:

a) $\frac{1}{2} \times 10^4$ b) $10^4\ \Omega$ c) $2 \times 10^4\ \Omega$ d) $25\ \Omega$

Q4 The resonant frequency for the given circuit will be:



a) 3 rad/sec b) 1 rad/sec c) 2 rad/sec d) 4 rad/sec

Q5 A series R-L-C circuit has a supply voltage of 100 V (variable frequency source), with a resistance of 10 Ω and inductive reactance of 50 Ω at 100 Hz. If resonant frequency is 500 Hz, then the voltage across the capacitor at resonance is:

a) 100 V b) 500 V c) 2500 V d) 5000 V

Q6 A series R-L-C circuit is in resonance at 100 kHz and has a quality factor of 50. The lower half frequency is:

a) 98 kHz b) 99 kHz c) 49 kHz d) 49.5 kHz

Q7 An R-L-C resonance circuit has a resonant frequency of 1.5 MHz and a bandwidth of 10 kHz. If C = 150 pF, then the effective resistance (in Ohms) of the circuit will be:

a) 29.5 b) 14.75 c) 9.4 d) 4.7

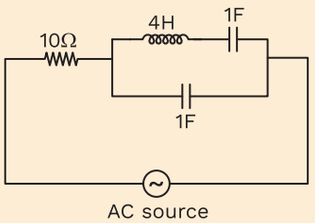
Q8 A series R-L-C circuit has a quality factor of 400, and an impedance of (100 + j50) Ω and its resonant angular frequency is 10 rad/sec. The value of R and L are, respectively:

a) 100 Ω, 10^{-3}H b) 10 Ω, 10^{2}H c) 1000 Ω, 10 H d) 100 Ω, 100 H

Q9 A series R-L-C circuit is supplied by a voltage of 220 V, 50 Hz. At resonance, the voltage across capacitor = 550 V, I = 1 A. Determine R, C, and L, respectively.

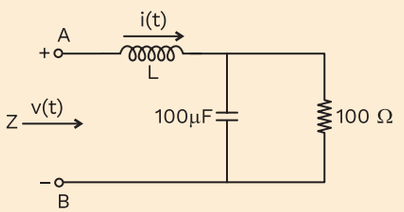
a) R = 110 Ω, C = 3.63 μF, L = 1.75 H
b) R = 220 Ω, C = 3.63 μF, L = 0.75 H
c) R = 220 Ω, C = 5.75 μF, L = 1.75 H
d) R = 110 Ω, C = 5.75 μF, L = 0.75 H

Q10 **The circuit shown below resonates at:**



a) All frequencies **b) 0.5 rad/sec** **c) 5 rad/sec** **d) 1 rad/sec**

Q11 **The voltage v(t) across the terminals A and B as shown in the figure, is a sinusoidal voltage having a frequency ω = 100 radian/sec. When the inductor current i(t) is in phase with the voltage v(t), the magnitude of the impedance Z(in Ω) seen between the terminals A and B is _____. (up to 2 decimal places)**



Answer Key:

1)	**b)**	**2)**	**c)**	**3)**	**d)**	**4)**	**a)**	**5)**	**c)**
6)	**b)**	**7)**	**d)**	**8)**	**a)**	**9)**	**c)**	**10)**	**b)**
11)	**50**								